

Application of STEM Principles and Practices:

The proposed solution is a handle for a cart, custom-tailored specifically for Mr. Kiser. His height and persistent back problems, caused by the need to bend down to reach the original handle, prompted the development of this innovative design. A standout feature of this handle is its adjustability, which allows it to be modified to precisely fit Mr. Kiser's height. Doing so, significantly reduces his need to bend and therefore helps alleviate his back strain. The design isn't just functional; it's also ergonomic, ensuring that Mr. Kiser finds it comfortable and easy to use, while engineering principles have been meticulously applied to guarantee both the handle's stability and durability.

The design of the adjustable handle takes into account several crucial factors. First and foremost, the materials chosen for its construction are selected for their strength and lightweight properties. This careful selection ensures that the handle is not only robust and capable of withstanding regular use but also remains easy to maneuver without causing additional strain. Furthermore, the adjustability mechanism has been designed with user-friendliness in mind. This ensures that Mr. Kiser can effortlessly modify the handle's height without the need for significant effort or technical expertise. This ease of adjustment means that the handle can be quickly set to the optimal height, offering immediate relief from back strain and enhancing Mr. Kiser's overall comfort.

In addition, the ergonomic design ensures that the handle's grip is comfortable, reducing the likelihood of hand fatigue or discomfort during extended use. The design also includes safety features to prevent the handle from slipping or collapsing unexpectedly, thereby providing Mr. Kiser with added peace of mind. Engineering principles ensure that the handle remains stable and durable, even under the stresses of daily use, while maintaining a sleek and modern aesthetic.

Moreover, the design process involved extensive testing and prototyping to ensure that the handle meets all of Mr. Kiser's specific needs. Feedback from initial tests was used to refine the design, addressing any potential issues and ensuring that the final product is both functional and reliable. By considering all these factors, the adjustable handle offers a comprehensive solution that not only addresses Mr. Kiser's immediate back problems but also enhances his overall experience with the cart, making it more user-friendly and comfortable for long-term use.